

Frank Angelo Malubag

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Summary

Backend-focused Software Engineer building production web applications, distributed systems, and cloud-native infrastructure. Experienced with TypeScript, Node.js, Go, Rust, PostgreSQL, Docker, Terraform, AWS, and event-driven architectures. Focused on backend engineering, system design, observability, and scalable distributed systems, continuously developing expertise through hands-on projects and technical literature.

Work Experience

ATLAS ANALYTICS

Software Developer

Singapore (Remote)

Nov 2023 - Dec 2023

- Hardened API security by shipping a stateless JWT strategy with token expiry and rotation, replacing session-based auth and reducing unauthorized access surface area across all SaaS platform endpoints.
- Engineered a report generation workflow aggregating user-specific analytics through Xano APIs and rendering them in WeWeb, enabling automated PDF-ready report generation.

Project Experience

Video Streaming Microservices Platform — 2026

github.com/Frankuccino/video-streaming-microservices

Node.js · RabbitMQ · Docker · Docker Compose · Terraform · AWS (EC2, ECR) · K3s

- Decomposed a monolithic video service into **4 independently deployable microservices**, replacing point-to-point HTTP calls with a **RabbitMQ fanout exchange** so 2 downstream consumers (history, recommendations) **process events concurrently** without modifying the publisher.
- Maintained separate dev/prod **Dockerfiles with hot-reload in development** and optimized multi-stage production builds ensuring **full environment parity** from first commit through deployment.
- **Provisioned AWS infrastructure via Terraform** (ECR, EC2) and configured a **self-managed K3s cluster** for Kubernetes-style orchestration of containerized workloads.

HR Management System — 2023

hrms-au39.onrender.com

Node.js · Express.js · MongoDB · Passport.js · MVC · Render

- Delivered a self-hosted and actively used **42-endpoint full-stack system** to ~5 admin staff who manage ~100 employee records in a local municipality, **supporting real-world personnel management workflows**.
- Replaced client-side access restrictions with **server-enforced authentication using Passport.js**, ensuring **protected resources remained secure** regardless of client behavior.
- Led backend architecture, MongoDB schema design, API development, deployment, and production support **from requirements gathering through rollout**.

Northstar Employee Management Platform — 2026

github.com/Frankuccino/northstar

TypeScript · React.js · TanStack Query · Node.js · Express.js · Vitest · Supertest · PostgreSQL

- Engineered **7 RESTful API endpoints** backed by **31 automated tests** covering success, validation, and edge-case scenarios.
- Hardened **API security** using **rate limiting** capped at 100 requests per 15-minute window, **Helmet** to enable security headers, and **CORS policies to reduce common web attack vectors**.
- Structured the codebase around a **feature-driven architecture** (Bulletproof React) with modular backend patterns and strict **separation of concerns**, keeping each layer independently testable and contained to prevent regressions across unrelated features.

Zero To Production Newsletter Service (Rust) — 2026

github.com/Frankuccino/rust-zero2prod

Rust · Actix Web · SQLx · PostgreSQL · Tokio · Tracing

- Built a **type-safe /subscriptions endpoint in Rust** with async PostgreSQL persistence via SQLx and **compile-time query validation** to leverage Rust's ownership model to ensure thread-safe request handling.
- Validated correctness with **2 black-box integration tests** covering the full request lifecycle: happy path and 400 on missing required fields.
- Instrumented **request-level telemetry** using the `tracing` crate to **improve observability and debugging**.

Skills

Languages: JavaScript, TypeScript, Go, Rust, SQL

Frontend: React, Next.js, HTML, CSS, TailwindCSS

Backend: RESTful APIs, Node.js, Express.js, PostgreSQL, MongoDB, SQLx, Actix Web

Cloud & Infrastructure: Docker, Docker Compose, Terraform, AWS, Cloudflare R2, K3s, RabbitMQ, GitHub Actions

Engineering Practices: Testing, CI/CD, Security Hardening, Observability, Git, Agile

Education

KodeGo

Full Stack Web Development Bootcamp

Aug 2023

St. Paul University Dumaguete

STEM

Dec 2021